K Nearest Neighbors (Continued)

★ Steps for KNN :-

1. Consider a query point & all the datapoints.
2. Calculate the distance b/w the query point & all the datapoints
3. Sort the distances
4. Select top-K distances
5. Look for majority voting
6. Class Assignment

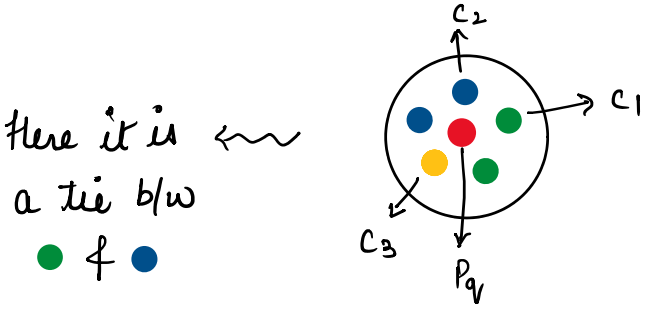
★ Weighted KNN :-

If there comes a tie b/w the two classes, then in such cases we go for weighted KNN.

formula → $w = \frac{1}{d}$

where, w = weight
d = distance

Ex- Suppose there are 3 classes

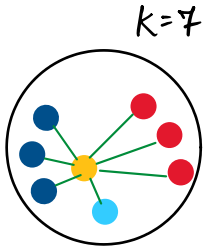


Suppose we calculated the distances

	distance	weight
●	1	1

K=7

	distance	weight
●	1	1
●	4	0.25
●	2	0.5
●	3	0.33
●	5	0.2



Here, we add the weights

● ⇝ 1 + 0.25 = 1.25 ⇝ High weight

● ⇝ 0.5 + 0.33 = 0.83

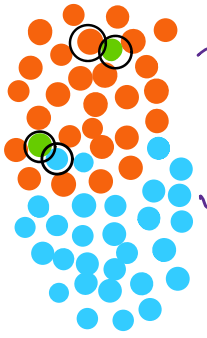
● ⇝ 0.2

∴ Pq will belong to ● class

NOTE:- Even after finding the weighted distance, there comes a tie, then we select the prediction randomly.

★ How to select the value of K?

I. K=1



⇝ for correct neighbors, there will be correct prediction

⇝ for incorrect neighbors, there will be wrong prediction

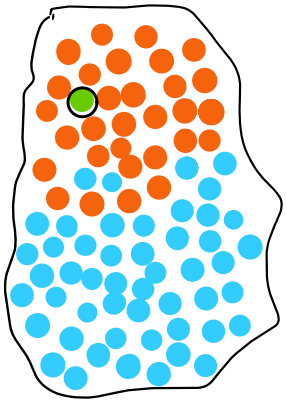
∴ we do not select a very small value of K. This condition is known as Overfitting

Train Data ⇝ Model performs well

↳

Train Data	⇝ Model performs well
Test Data	⇝ " performs poor

II. K=n



⇝ Acc. to K=n, majority class will be blue so, we say query point belongs to blue class which is incorrect prediction.

∴ This leads to the problem of underfitting

↓

Train Data	⇝ Model performs poor
Test Data	⇝ " performs poor

So, we can say that →

if K ↓↓ ⇝ overfitting

K ↑↑ ⇝ underfitting

So, for perfectly fit model, we select a moderate value of K

Thus K acts as a hyperparameter.

★ Types of distance calculation metrics :-

① Euclidean Distance :- Continous, low-dim. data

$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + \dots}$$

② Manhattan Distance :- Categorical, Normal dimension data

$$d = |x_2-x_1| + |y_2-y_1| + \dots$$

③ Cosine Distance :- Very high-dimensional data

$$\cos\theta = \frac{A \cdot B}{|A| \cdot |B|}$$

④ Minkowski Distance :- Categorical, high-dimension data

$$d = \sqrt[p]{(x_2-x_1)^p + (y_2-y_1)^p + \dots}$$

* Performance Metrics :-

Accuracy :- It is a metric used for checking the performance of the model in case of classification problem.

$$\text{Accuracy} = \frac{\text{No. of correct predictions}}{\text{Total No. of predictions}}$$

Ex- Total unknown points (n) = 1000

Correct Predictions = 820

Incorrect " = 180

$$\therefore \quad \text{Acc} = \frac{820}{1000} = 0.82 \text{ or } 82\%$$

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